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Measurement of Age,Distance and Metallicity of Globular Cluster NGC 7089 using Color-Magnitude Diagram

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ABSTRACT

In this study we investigate the color-magnitude diagram of globular cluster NGC 7089 to determine its distance, age and metallicity. For this purpose we take images of the cluster using 20 inch telescope at HRPO in different bands. We analyze the images by applying different python packages to apply photometry and calibration and extract the magnitudes of the individual stars of the cluster. By using the magnitudes we create the color-magnitude diagram. We compare the color-magnitude diagrams with range of isochrones allowing age, metallicity and distance to be measured with reasonable uncertainty.

Keywords: Globular Cluster, CMD, metallicity, HRPO, isochrones, photometry

1. INTRODUCTION

Globular clusters (GCs) are among the oldest objects in the universe and accompany most major star formation episodes in galaxies (Brodie & Strader 2006). They are among the oldest objects in their galaxies and even the universe, constraining estimates of the universe's age. Since globular clusters form at once from a single giant molecular cloud, a cluster's stars have roughly the same age and composition. Stars in a cluster formed almost simultaneously from the same cloud of interstellar gas, which means that the stars in the cluster should be very homogeneous in their properties. This means if we measure the properties the cluster (age, distance, composition, etc.) in general, we can assume that the properties of the stars in the cluster will be very similar. To determine these properties we need to construct the Color-Magnitude Diagram for the cluster from the magnitudes of the observed stars of the cluster. A Color-Magnitude Diagram (CMD) shows us the plot of brightness (or luminosity) against color (surface temperature) of the stars. The plot of luminosity against temperature is also known as Hertzsprung-Russell (or H-R) diagram (Hertzsprung 1906; Russell 1914).CMD is a essential tool for classifying stellar types and their properties. Using CMD we can calculate the properties like distance, age and metallicity of a globular cluster.
The globular cluster chosen for this study is NGC 7089 (also designated Messier 2 or M2). Situated in the constellation Aquarius NGC 7089 is one of the largest known globular clusters. This globular cluster was chosen based on the during the period of observations. The goal of this study is to determine the distance, age and metallicity of this cluster. For this purpose images of the cluster was analyzed using various python packages and CMD was constructed. Then the CMD was fitted with isochrones for calculating the desired parameters.

2. OBSERVATIONS

The observations for this study was done using the HRPO 20 inch telescope. For observation of globular clusters the signal to noise ratio need to be 10(Sarajedini 2007). Using this S/N we approximated the required observing time. During the observation period we were able to get 16 images of the target with 60s exposure in each band.

3. DATA ANALYSIS

The images of the cluster taken from the telescope was reduced, plate solved and stacked using different python pipelines. At this point we have 3 images of the cluster in 3 different filters. We need to extract the magnitudes of individual stars and calibrate the magnitudes. For this purpose we used photometry and calibration.

3.1. *Photometry*

The globular cluster is very crowded field with a huge number of stars. For this reason the regular aperture photometry pipeline could not be used. We used Point Spread Function Photometry(PSF) which is a part of python photoutils package. In this method we build a effective PSF using few bright isolated stars from our image, and used the IterativelySubtractedPSFPhotometry to get the photometric data. The photometry process gave us the v and r magnitudes of the stars along with errors which were saved in data files.

3.2. *Calibration*

For calibration I selected a magnitude of star from my data and magnitude of the same star from Gaia Survey data and calculated the difference between the magnitudes. Then I added the difference to all the magnitudes. And this was done for both v and r band magnitudes. This very rudimentary and rough way for calibration was chosen instead of the standard way, because of time constrains. The magnitude of the chosen star from gaia survey data was calculated using the equation,

$$R = Gmag + 0.003226 - 0.3833 * (BP - RP) + 0.1346 * (BP - RP)^2 \quad (1)$$

$$V = Gmag + 0.01760 + 0.006860 * (BP - RP) + 0.1732 * (BP - RP)^2 \quad (2)$$

Here, Gmag, BP, and RP are Gaia magnitudes (in the Gaia catalog).

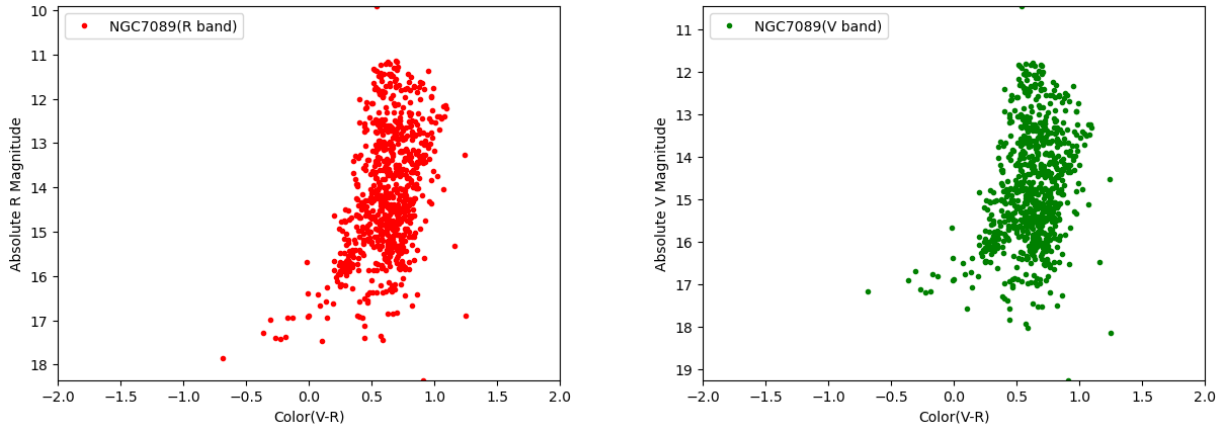


Figure 1. Color-Magnitude Diagram for NGC 7089 in R and V Band

3.3. *Isochrone Fitting*

Using the calibrated magnitudes we plotted the CMD (figure 1) for NGC 7089 for both V and R bands. To measure the cluster parameters we need to apply isochrone fitting to our CMD diagrams. First we need to get isochrone data or a large range of age and metallicity. The isochrone data were downloaded using ezPADOVA-2, a python package used to get PARSEC isochrones from the CMD website and format them to be used with ASteCA. We did not use ASteCA for the plotting the isochores, instead used our own python scripts. The isochrones used had age range of 10^5 to $10^{10.5}$ years and metallicity range of -2.2 to 4.4 dex. A suitable sub-range of isochrones where chosen and

plotted on top of the CMD from collected data. Isochrone data also had to be converted to V and R magnitudes using equation (1) and (2). By trial and error I went through the whole range of isochrone data and a best fit sub-range was chosen whose shape best fits the color-magnitude diagrams. This fitting gives us the range of age and metallicity for the cluster. The distance of the cluster is measured using spectroscopic parallax (Carol & Ostile 2007).The isochrone data is vertically shifted to fit the CMD, this vertical shift is the distance modulus(m-M). Using the distance modulus the cluster distance was calculated using the equation,

$$d = 10^{\frac{m-M+5}{5}} \tag{3}$$

Thspectroscopic parallax The average of the range of age and metallicity gives us the age and metallicity of the cluster and the range serves as the uncertainty. While using the disatnce modulus we calculated the distance. The results are discussed in the following section.

3.4. *Error Analysis*

In case of CMD we need the standard deviations of the magnitudes in both bands. In the photometry section, the PSF photometry function gives us the errors in magnitude as output. In the calibration steps the gaia survey data contains the errors in Gmag, BP, and RP magnitudes. These were used to calculate the errors in V and R magnitudes for the chosen star from gaia survey.

$$\sigma_{BP-RP} = \sqrt{\sigma_{BP}^2 + \sigma_{RP}^2} \tag{4}$$

$$\sigma_{Gaia-R} = \sqrt{\sigma_{Gmag}^2 + (-0.3833 + 2 * 0.1346 * (BP - RP))^2 \sigma_{BP-RP}^2} \tag{5}$$

$$\sigma_{Gaia-V} = \sqrt{\sigma_{Gmag}^2 + (0.006860 + 2 * 0.1732 * (BP - RP))^2 \sigma_{BP-RP}^2} \tag{6}$$

Then we calculated the calibration factor which is the difference between the gaia magnitude and the observed magnitude of the chosen star. The error in calibration factor for both bands,

$$\sigma_{C_{factor_R}} = \sqrt{\sigma_{Gaia-R}^2 + \sigma_R^2} \tag{7}$$

$$\sigma_{C_{factor_V}} = \sqrt{\sigma_{Gaia-V}^2 + \sigma_V^2} \tag{8}$$

Here σ_R and σ_V are the observed magnitudes of the stars.

Finally the standard deviations of the calibrated magnitudes were calculated.

$$\sigma_R = \sqrt{\sigma_{C_{factor_R}}^2 + \sigma_R^2} \tag{9}$$

$$\sigma_V = \sqrt{\sigma_{C_{factor_V}}^2 + \sigma_V^2} \tag{10}$$

The standard deviations for color (V-R),

$$\sigma_{V-R} = \sqrt{\sigma_V^2 + \sigma_R^2} \tag{11}$$

The errors in magnitudes were calculated to be used as error bars for CMD, but unfortunately the errors calculated were too large, if plotted they span the whole graph. The reason for this large errors were the PSF photometry package. The errors output from PSF photometry function were too large, possibly because of improper use of the function or faulty input parameters.

4. RESULTS

4.1. *Color-Magnitude Diagrams*

Figure 1 shows the CMD for NGC 7089 in R and V band. As we can see the color magnitude diagram does not quite represent a typical CMD for a cluster. We do not observe the main sequence part, as main sequence stars are bright and it would require better telescope and large observing time. We do see a horizontal branch which represents the supergiant stars. The shape of the CMD is not typical, which indicates the photometry method may not have worked as desired. Further investigation is required to fix the issue.

4.2. *Isochrones*

Figure 2 shows the CMD fitted with isochrones with specific ages and metallicities.

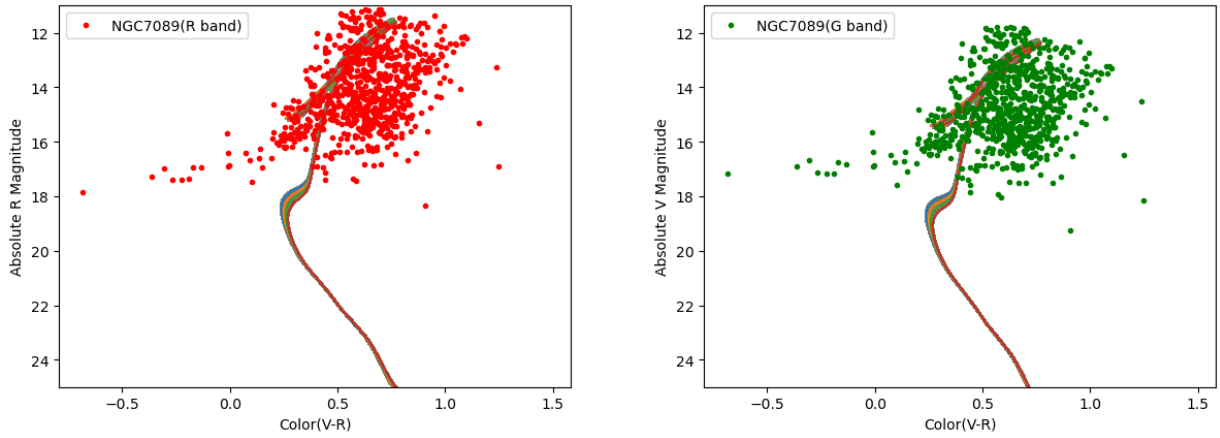


Figure 2. Color-Magnitude Diagram for NGC 7089 with isochrone fitting

The distance modulus, the vertical shift of the isochorones is $m - M = 15$. Using equation (3) the distance to the cluster was calculate to be 10,000 parsecs. The age range for the fitted isochrones is $10 - 15.8$ billion years. And the matallicity range for the fitted isochrones -1.7 to -1.6 dex. The results are tabulated below along with accepted values.

Table 1. Results

Column	Distance (Kiloparsecs)	Age (Billion Years)	Metallicity (dex)
Observed Values	10	10 15	-1.7 -1.6
Accepted values	17	13	-1.65

From the table 1 we can see that the observed parameters are fau=irly close to the actual values.

5. CONCLUSION

In this project I have attempted to calculate the parameters of globular cluster NGC 7089 by analysing the CMD of the stars of the clusters by applying isochrone fitting. Although the results seem fairly close to acti=ula values, the CMDs themselves do not seem to be exactly right. They are very broad, which is why a very wide range of isochrones seem to fit. I attempted to fit the isochrones using the horizontal branch. The reason for the broad CMD and also the unsuccessful error analysis seemed to be the PSF photometry. The photometry outputs seemed to be faulty. This may be caused by improper use of the PSF function or inappropriate inputs. Overall the results seems to indicate that it is possible to measure various properties of a globular cluster using CMDs created from data obtained from a ground based 20 inch telescope.

6. SOFTWARE AND THIRD PARTY DATA REPOSITORY CITATIONS

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